WELL-COVERED GRAPH POWERS CHARACTERIZED IN CHORDAL, VERY WELL-COVERED AND COCHORDAL CLASSES

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Editorial note. The chordal, very well-covered and cochordal classification statements and their full proofs are one bundled structural object; clique-blow-up invariance and other corollaries remain complete context rather than extra counts.

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Allowed standard prerequisites (editorial).

- Well-covered graph structures and independence complexes. All original definitions, exact graph-class/connectedness/isolated-vertex and field-characteristic hypotheses, original supporting statements and citations are retained with complete source-local arguments.. Full author paper context and native bibliography supplied; no proof reconstruction, assumption repair or independent certification.

1. Introduction

Let G be a simple graph. A subset $C \subseteq V(G)$ is called a *vertex cover* of G if C intersects every edge of G. It is called a *minimal vertex cover* if no proper subset of C is a vertex cover. A graph G is said to be *well-covered* if all of its minimal vertex covers have the same cardinality.

The concept of well-coveredness is a central topic in graph theory [BM, P]. Structural characterizations of well-covered graphs have been established for several prominent graph classes, including chordal graphs [PTV], bipartite graphs [R], and graphs with girth$(G) \geq 5$ [FHN]. On the other hand, graph powers constitute another fundamental notion in structural graph theory. For a positive integer d, the d-th power G^d is defined on the vertex set $V(G^d) = V(G)$, where two distinct vertices $u, v \in V(G)$ form an edge $\{u, v\} \in E(G^d)$ if and only if they are connected by a path of length at most d in G. In this work, we study the following classification problem:

Problem 1.1. *Classify all simple graphs G for which G^d is well-covered for every $d \geq 1$.*

In the bipartite setting, Deniz [D] showed that if both G and G^2 are well-covered, then every connected component of G is isomorphic to either K_1 or $K_{n,n}$. This suggests that requiring G^d to be well-covered for all $d \geq 1$ imposes severe structural constraints on G. Since a graph G is well-covered if and only if each of its connected components is well-covered, we may assume without loss of generality that G is connected and non-trivial (i.e., has no isolated vertices).

Our first main results classify such graphs within two important classes: chordal graphs and very well-covered graphs.

Theorem 1.2. *Let G be a connected chordal graph with no isolated vertices. Then the following conditions are equivalent:*

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(1) G^d is well-covered for all $d \geq 1$;
(2) G and G^2 are well-covered;
(3) G is a complete graph K_n for some $n \geq 2$.

Theorem 1.3. *Let G be a connected, very well-covered graph with no isolated vertices. Then the following conditions are equivalent:*

(1) G^d is well-covered for all $d \geq 1$;
(2) G and G^2 are well-covered;
(3) G is isomorphic to $K_{n,n}$ for some $n \geq 1$.

Recall that for a connected graph G, the distance $d_G(u, v)$ between two vertices $u, v \in V(G)$ is the length of a shortest path connecting them. The *diameter* of G, denoted by $\operatorname{diam}(G)$, is defined as

$$\operatorname{diam}(G) = \max\{d_G(u, v) \mid u, v \in V(G)\}.$$

Notice that if $\operatorname{diam}(G) \leq 2$, then G^2 (and indeed G^d for all $d \geq 2$) is a complete graph K_n, which is trivially well-covered. Our next main result demonstrates that for cochordal graphs, this diameter condition completely characterizes when all powers remain well-covered.

Theorem 1.4. *Let G be a connected cochordal graph with no isolated vertices. Then the following conditions are equivalent:*

(1) G^d is well-covered for all $d \geq 1$;
(2) G and G^2 are well-covered;
(3) G is well-covered and $\operatorname{diam}(G) \leq 2$.

Furthermore, we demonstrate that this property is closed under clique blow-ups (clique substitutions). Recall the definition of graph composition:

Definition 1.5. Let G be a simple graph with vertex set $V(G) = \{1, \ldots, n\}$, and let $H_1, \ldots, H_n$ be graphs. The *composition graph* $G[H_1, \ldots, H_n]$ is obtained by replacing each vertex $i \in V(G)$ with the graph H_i, and joining every vertex of H_i to every vertex of H_j whenever $\{i, j\} \in E(G)$.

We show that replacing vertices with complete graphs preserves the property that all powers remain well-covered:

Theorem 1.6. *Let G be a simple graph with $V(G) = \{v_1, \ldots, v_n\}$. For any positive integers $a_1, \ldots, a_n$, let $H = G[K_{a_1}, \ldots, K_{a_n}]$ be the clique blow-up of G. Then G^d is well-covered for all $d \geq 1$ if and only if H^d is well-covered for all $d \geq 1$.*

The paper is organized as follows. In Section 2, we collect necessary notation, definitions, and preliminary results on well-covered graphs and graph powers, and establish several basic properties of graphs whose powers are all well-covered. Section 3 is devoted to proving our main classification results for chordal, very well-covered, and cochordal graphs, as well as establishing stability under clique blow-ups.

2. Graphs, Powers, and Well-coveredness

2.1. Graphs.

Definition 2.1. Let $G = (V(G), E(G))$ be a simple graph on the vertex set $V(G) = \{1, \ldots, n\}$.

(1) The *neighborhood* of a vertex $x \in V(G)$ is $N_G(x) = \{y \in V(G) \mid \{x, y\} \in E(G)\}$. The *closed neighborhood* of x is $N_G[x] = N_G(x) \cup \{x\}$.
(2) For a subset $U \subseteq V(G)$, the *neighborhood* and *closed neighborhood* of U in G are defined respectively by

$$N_G(U) = \bigcup_{x \in U} N_G(x) \quad \text{and} \quad N_G[U] = \bigcup_{x \in U} N_G[x].$$

(3) A graph H is a *subgraph* of G if $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$. It is an *induced subgraph* of G if $E(H) = \{\{u, v\} \in E(G) \mid u, v \in V(H)\}$. For a subset $U \subseteq V(G)$, $G[U]$ denotes the induced subgraph of G on U, and $G \setminus U$ denotes $G[V(G) \setminus U]$.
(4) A *path* P_k on k vertices has vertex set $V(P_k) = \{v_1, \ldots, v_k\}$ and edge set $E(P_k) = \{\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{k-1},$
A *cycle* C_k on k vertices ($k \geq 3$) has edge set $E(C_k) = E(P_k) \cup \{\{v_1, v_k\}\}$.
(5) A graph G is *chordal* if it contains no induced cycle of length four or greater. A graph G is *cochordal* if its complement graph $\overline{G}$ is chordal.

Definition 2.2 (Distance and Graph Powers)**.** Let G be a connected graph. The *distance* $d_G(u, v)$ between two vertices $u, v \in V(G)$ is the length of a shortest path in G connecting them. The *diameter* of G is $\operatorname{diam}(G) = \max\{d_G(u, v) \mid u, v \in V(G)\}$.

For a positive integer $d \geq 1$, the *d-th power* of G, denoted by G^d, is the graph defined on the vertex set $V(G^d) = V(G)$ with edge set

$$E(G^d) = \{\{u, v\} \subseteq V(G) \mid u \neq v \text{ and } d_G(u, v) \leq d\}.$$

Definition 2.3 (Independent Sets, Vertex Covers, and Well-Coveredness)**.** Let G be a simple graph.

(1) A subset $I \subseteq V(G)$ is an *independent set* of G if no two vertices in I are adjacent in G. An independent set I is *maximal* if it is not properly contained in any other independent set of G. The *independence number* of G, denoted by $\alpha(G)$, is the cardinality of a maximum independent set of G.
(2) A subset $C \subseteq V(G)$ is a *vertex cover* of G if $C \cap e \neq \emptyset$ for every $e \in E(G)$. A vertex cover C is *minimal* if no proper subset of C is a vertex cover of G. Note that $I \subseteq V(G)$ is a maximal independent set of G if and only if $V(G) \setminus I$ is a minimal vertex cover of G.
(3) A graph G is *well-covered* if every minimal vertex cover of G has the same cardinality.

Definition 2.4. Let G be a simple graph and v a vertex of G. The *eccentricity* of v is defined by $\operatorname{ecc}_G(v) = \max\{d_G(u, v) \mid u \in V(G)\}$. The *radius* of G is defined by $\operatorname{rad}(G) = \min\{\operatorname{ecc}_G(v) \mid v \in V(G)\}$. A graph G is called *self-centered* if $\operatorname{rad}(G) = \operatorname{diam}(G)$.

Lemma 2.5. *Let G be a simple connected graph of diameter D. Assume that G^{D-1} is well-covered. Then every vertex v of G has eccentricity D. In other words, G is self-centered.*

Proof. Assume for contradiction that $\operatorname{ecc}_G(v) < D$. Then $N_{G^{D-1}}[v] = V(G)$. Hence, the maximal independent set of G^{D-1} containing v is $\{v\}$, which has size 1. However, since $\operatorname{diam}(G) = D$, there exist vertices $u, w \in V(G)$ such that $d_G(u, w) = D > D - 1$, meaning u and w are independent in G^{D-1}. Thus, any maximal independent set containing $\{u, w\}$ has size at least 2, contradicting the assumption that G^{D-1} is well-covered. The conclusion follows. □

Corollary 2.6. *Let G be a simple connected graph of diameter D. Assume that $D \geq 2$ and G^{D-1} is well-covered. Then G has no leaves.*

Proof. Suppose for the sake of contradiction that G contains a leaf v with unique neighbor u. Then for every vertex $w \notin \{u, v\}$, we have $d_G(v, w) = 1 + d_G(u, w)$. Consequently, $\operatorname{ecc}_G(v) = \operatorname{ecc}_G(u) + 1$, which contradicts Lemma 2.5. The conclusion follows. □

Let G be a simple graph and $v \in V(G)$ a vertex. For a non-negative integer $a \geq 0$, the *closed ball* of radius a centered at v is defined as

$$B_a(v) = \{u \in V(G) \mid d_G(u, v) \leq a\}.$$

Note that $B_d(v) = N_{G^d}[v]$ is precisely the closed neighborhood of v in G^d.

Lemma 2.7. *Let G be a simple graph such that G^d is well-covered for some $d \geq 1$. Then there exist no vertices $s, t, w \in V(G)$ such that $d_G(s, t) \geq d + 1$ and $B_d(s) \cup B_d(t) \subseteq B_d(w)$.*

Proof. Suppose for the sake of contradiction that there exist vertices $s, t, w \in V(G)$ satisfying $d_G(s,t) \geq d+1$ and $B_d(s) \cup B_d(t) \subseteq B_d(w)$. Let X be a maximal independent set of G^d containing w. Since G^d is well-covered, we have $|X| = \alpha(G^d)$.

By definition of independent sets in graph powers, no vertex in $X \setminus \{w\}$ can belong to $N_{G^d}[w] = B_d(w)$. Since $B_d(s) \cup B_d(t) \subseteq B_d(w)$, it follows that

$$(X \setminus \{w\}) \cap (B_d(s) \cup B_d(t)) = \emptyset.$$

Furthermore, because $d_G(s,t) \geq d+1$, $\{s,t\}$ is an independent set in G^d. Consequently, the set $Y = (X \setminus \{w\}) \cup \{s,t\}$ is an independent set of G^d. However,

$$|Y| = |X| - 1 + 2 = |X| + 1 = \alpha(G^d) + 1,$$

which contradicts the assumption that $\alpha(G^d)$ is the maximum size of an independent set in G^d. Thus, no such vertices s, t, w can exist. □

3. Well-coveredness of graph powers

3.1. **Chordal Graphs.** We begin by recalling several fundamental concepts in chordal graph theory.

Definition 3.1. Let G be a graph. A vertex $v \in V(G)$ is called a *simplicial vertex* if its closed neighborhood $N_G[v]$ induces a clique in G. A clique $K \subseteq V(G)$ is called a *simplex* of G if $K = N_G[v]$ for some simplicial vertex $v \in V(G)$.

We have the following fundamental theorem due to Prisner, Topp, and Vestergaard [PTV]:

Theorem 3.2 (Well-covered chordal graphs)**.** *A chordal graph G is well-covered if and only if every vertex of G lies in exactly one simplex.*

Proof of Theorem 1.2. The implications (1) $\implies$ (2) and (3) $\implies$ (1) are obvious. It remains to prove that if G and G^2 are well-covered, then G is a complete graph.

By Theorem 3.2, since G is a well-covered chordal graph, its vertices are partitioned into pairwise disjoint simplices $S_1, \ldots, S_m$. For each $1 \leq i \leq m$, let $s_i \in S_i$ be a simplicial vertex. Then $\{s_1, \ldots, s_m\}$ is an independent set in G^2 because $N_G[s_i] = S_i$ and $S_i \cap S_j = \emptyset$ for all $i \neq j$, meaning $d_G(s_i, s_j) \geq 3$. Moreover, every vertex in G belongs to some $S_i = N_G[s_i]$ and is thus within distance 1 or 2 from s_i. Therefore, $\{s_1, \ldots, s_m\}$ is a maximal independent set of G^2, so $\alpha(G^2) = m$.

Suppose for the sake of contradiction that G is not a complete graph, so $m \geq 2$ and there exists a non-simplicial vertex $v \in V(G)$. Without loss of generality, assume $v \in S_a$. Since v is non-simplicial, $N_G[v]$ is not a clique, so v must have a neighbor w outside S_a; say $w \in S_b$ for some $b \neq a$. Since $S_a = N_G[s_a]$ and $S_b = N_G[s_b]$ are cliques, every vertex in $S_a \cup S_b$ is within distance at most 2 from v in G, so $S_a \cup S_b \subseteq N_{G^2}[v]$.

Consequently, any maximal independent set X of G^2 containing v cannot contain any vertices from $S_a \cup S_b$. Since X can pick at most one vertex from each remaining simplex S_k ($k \notin \{a, b\}$), we obtain $|X| \leq m - 1 < \alpha(G^2)$, contradicting the assumption that G^2 is well-covered. Thus, G contains no non-simplicial vertices, so $m = 1$ and G is a complete graph. □

3.2. **Very well-covered graphs.** We recall the following characterization of very well-covered graphs due to Favaron [F, Theorem 1.2].

Theorem 3.3 (Very well-covered graphs)**.** *A graph G with a perfect matching M is very well-covered if and only if every edge $\{u, u'\} \in M$ satisfies the following conditions:*

(1) $N_G(u) \cap N_G(u') = \emptyset$;
(2) every vertex $x \in N_G(u) \setminus \{u'\}$ is adjacent to every vertex $y \in N_G(u') \setminus \{u\}$.

Proof of Theorem 1.3. The implications (1) $\implies$ (2) and (3) $\implies$ (1) are obvious. It remains to prove that if G is very well-covered and G^2 is well-covered, then G is a complete bipartite graph.

Let M be a perfect matching of G satisfying the conditions of Theorem 3.3. For each vertex $x \in V(G)$, denote by x' its unique matched partner under M.

Step 1: *Construction of neighborhood sets S and T.*

Let $u \in V(G)$ be a vertex of minimum degree in G. Set $S = N_G(u)$ and $T = N_G(u')$.

For any vertex $x \in S = N_G(u)$, applying condition (2) of Theorem 3.3 to the matching edge $\{x, x'\}$ with $u \in N_G(x)$ yields $N_G(x') \setminus \{x\} \subseteq N_G(u)$. Since $x \in N_G(u)$, it follows that $N_G(x') \subseteq N_G(u)$. By the minimality of $\deg_G(u)$, we obtain $N_G(x') = N_G(u) = S$.

Next, applying condition (2) to the matching edge $\{u, u'\}$ with $x \in N_G(u)$ implies $N_G(u') \setminus \{u\} \subseteq N_G(x)$. Since $x \in N_G(u)$, we also have $u \in N_G(x)$, whence $N_G(u') \subseteq N_G(x)$. Conversely, applying condition (2) to the matching edge $\{x, x'\}$ with $u' \in N_G(x')$ gives $N_G(x) \subseteq N_G(u')$. Thus,

$$N_G(x) = N_G(u') = T \quad \text{for all } x \in S.$$

In particular, S and T are disjoint independent sets, and every vertex in S is adjacent to every vertex in T. Furthermore, for any $w \in T$, since $u' \in N_G(w)$, applying condition (2) to the matching edge $\{w, w'\}$ yields $N_G(w') \subseteq N_G(u') = T$.

Step 2: *If there exists $w \in T$ such that $w' \notin S$, then G^2 is not well-covered.*

Suppose there exists $w \in T$ such that $w' \notin S$. Since $w' \notin S = N_G(u)$ and $N_G(w') \subseteq T = N_G(u')$, the sequence (u, u', w, w') forms an induced path of length 3 in G, so $d_G(u, w') = 3$. We compare the closed 2-balls centered at u, u', and w':

$$\begin{aligned} B_2(u) &= \{u\} \cup S \cup T, \\ B_2(u') &= \{u'\} \cup T \cup N_G(T), \\ B_2(w') &= \{w'\} \cup N_G(w') \cup N_G(N_G(w')). \end{aligned}$$

Since $N_G(w') \subseteq T$, we have $N_G(N_G(w')) \subseteq N_G(T)$, which implies $B_2(w') \subseteq B_2(u')$. Moreover, $B_2(u) \subseteq B_2(u')$ clearly holds. Therefore,

$$B_2(u) \cup B_2(w') \subseteq B_2(u') \quad \text{with } d_G(u, w') \geq 3,$$

which contradicts Lemma 2.7. Hence, G^2 cannot be well-covered under this assumption.

Step 3: *If $w' \in S$ for all $w \in T$, then $G \cong K_{n,n}$.*

Suppose $w' \in S$ for all $w \in T$. Then for every $w \in T$, w' is a vertex in S, so $N_G(w') = T$.

Now assume there exists some $w \in T$ having a neighbor $z \notin S$. Since $w' \in S$, the pair $\{w, w'\}$ is a matching edge with $N_G(w') = T$. By condition (2) applied to $\{w, w'\}$, the neighbor $z \in N_G(w)$ must be adjacent to every vertex in $N_G(w') = T$. In particular, z is adjacent to u, which implies $z \in N_G(u) = S$, a contradiction.

Therefore, $N_G(w) = S$ for all $w \in T$, and $N_G(s) = T$ for all $s \in S$. Consequently, $S \cup T$ forms a connected component of G. Since G is connected, $V(G) = S \cup T$. Finally, since G is very well-covered, $|S| = |T| = n$, and thus $G \cong K_{n,n}$.

The conclusion follows. □

3.3. **Cochordal graphs.** The case of cochordal graphs will follow from the following lemma.

Lemma 3.4. *Let G be a connected cochordal graph. Then* $\mathrm{rad}(G) \leq 2$.

Proof. Let $H = \overline{G}$ be the complement graph of G. By definition, H is a chordal graph. Let s be a simplicial vertex of H, and let $Q = N_H[s]$ be the corresponding clique in H. By the definition of graph complementation, all vertices $u \notin Q$ are adjacent to s in G.

Now, let $t \in Q \setminus \{s\}$. Suppose for the sake of contradiction that $N_G(t) \subseteq Q$. This implies that $N_H[t] \supseteq V(G) \setminus Q$. Since Q is a clique in H, t is also adjacent in H to every vertex in $Q \setminus \{t\}$, so $N_H(t) \supseteq Q \setminus \{t\}$. Consequently, $N_H(t) = V(G) \setminus \{t\}$, which means t has no neighbors in G. Thus, t is an isolated vertex in G, contradicting the assumption that G is connected.

Therefore, every $t \in Q \setminus \{s\}$ must have a neighbor $z \notin Q$ in G. Since s is adjacent to z in G, the sequence (s, z, t) forms a path of length 2 in G, so $d_G(s,t) \leq 2$. Because s is also adjacent in G to every vertex outside Q, we obtain $\mathrm{ecc}_G(s) \leq 2$. It follows that $\mathrm{rad}(G) \leq \mathrm{ecc}_G(s) \leq 2$. □

Proof of Theorem 1.4. The implications (1) $\implies$ (2) and (3) $\implies$ (1) are obvious. The implication (2) $\implies$ (3) follows directly from Lemma 3.4 and Lemma 2.5. □

3.4. **Clique substitutions.**

Proof of Theorem 1.6. Let $V(G) = \{v_1, \ldots, v_n\}$. By definition of the clique blow-up, $H = G[K_{a_1}, \ldots, K_{a_n}]$ is obtained by replacing each vertex $v_i \in V(G)$ with a clique $Q_i \cong K_{a_i}$, where $a_i \geq 1$. Two distinct vertices $x \in Q_i$ and $y \in Q_j$ are adjacent in H if and only if $i = j$ or $v_i v_j \in E(G)$.

Define the canonical projection map $\pi : V(H) \to V(G)$ by setting $\pi(x) = v_i$ for all $x \in Q_i$.

Claim 1: *For any $x, y \in V(H)$ with $\pi(x) \neq \pi(y)$, we have $d_H(x,y) = d_G(\pi(x), \pi(y))$.*

Proof of Claim 1. Let $x \in Q_i$ and $y \in Q_j$ with $i \neq j$. If $v_i = u_0, u_1, \ldots, u_k = v_j$ is a path of length k in G, we can construct a path of length k in H from x to y by selecting an arbitrary representative $x_r \in Q_r$ for each $1 \leq r \leq k-1$ and setting $x_0 = x$ and $x_k = y$. Thus, $d_H(x,y) \leq d_G(v_i, v_j)$. Conversely, if $x = y_0, y_1, \ldots, y_m = y$ is a path of length m in H, then $\pi(y_0), \pi(y_1), \ldots, \pi(y_m)$ yields a walk (and hence contains a path) from v_i to v_j in G of length at most m. Therefore, $d_G(v_i, v_j) \leq d_H(x,y)$, establishing $d_H(x,y) = d_G(v_i, v_j)$. □

Fix an integer $d \geq 1$. By Claim 1, for any two distinct vertices $x \in Q_i$ and $y \in Q_j$, we have $d_H(x,y) \leq d$ if and only if $i = j$ or $d_G(v_i, v_j) \leq d$. In other words, two distinct vertices $x, y \in V(H)$ are adjacent in H^d if and only if either x, y belong to the same clique Q_i, or their projections $\pi(x), \pi(y)$ are adjacent in G^d.

Claim 2: *A subset $I \subseteq V(H)$ is a maximal independent set of H^d if and only if I contains exactly one vertex from Q_i for each $v_i \in J$, where $J = \pi(I)$ is a maximal independent set of G^d.*

Proof of Claim 2. Suppose I is a maximal independent set of H^d.

(1) Since every clique Q_i induces a clique in H^d, I can contain at most one vertex from each Q_i. Thus, the restriction of π to I is injective, so $|\pi(I)| = |I|$.
(2) For any two distinct vertices $x, y \in I$, we have $d_H(x,y) \geq d+1$. By Claim 1, $d_G(\pi(x), \pi(y)) = d_H(x,y) \geq d+1$, which implies that $J = \pi(I)$ is an independent set of G^d.
(3) To show J is maximal in G^d, suppose there exists $v_k \in V(G) \setminus J$ such that $J \cup \{v_k\}$ is independent in G^d. Pick an arbitrary vertex $z \in Q_k$. Then $d_G(v_k, \pi(x)) \geq d+1$ for all $x \in I$, so $d_H(z,x) \geq d+1$ for all $x \in I$. Hence $I \cup \{z\}$ is an independent set of H^d, contradicting the maximality of I. Thus, J is a maximal independent set of G^d.

Conversely, let J be a maximal independent set of G^d. For each $v_i \in J$, pick a vertex $x_i \in Q_i$, and set $I = \{x_i \mid v_i \in J\}$. By Claim 1, $d_H(x_i, x_j) = d_G(v_i, v_j) \geq d+1$ for all $i \neq j$, so I is independent in H^d. If I were not maximal in H^d, there would exist $y \in V(H) \setminus I$ such that $I \cup \{y\}$ is independent in H^d. If $y \in Q_i$ for some $v_i \in J$, then y is adjacent to $x_i \in I$ in $H \subseteq H^d$, a contradiction. If $y \in Q_k$ for some $v_k \notin J$, then $d_G(v_k, v_i) = d_H(y, x_i) \geq d+1$ for all $v_i \in J$, which implies $J \cup \{v_k\}$ is an independent set of G^d, contradicting the maximality of J. Therefore, I is a maximal independent set of H^d with $|I| = |J|$. □

By Claim 2, the cardinalities of all maximal independent sets of H^d are in one-to-one correspondence with the cardinalities of the maximal independent sets of G^d. Consequently, all maximal independent sets of H^d have the same cardinality if and only if all maximal independent sets of G^d have the same cardinality. Hence, G^d is well-covered for all $d \geq 1$ if and only if H^d is well-covered for all $d \geq 1$. □

Corollary 3.5. *Let G be a simple graph. Suppose $u, v \in V(G)$ are true twins in G, that is, $\{u, v\} \in E(G)$ and $N_G[u] = N_G[v]$. Let F be the graph obtained by contracting the edge $\{u, v\}$ to a single vertex. Then G^d is well-covered for all $d \geq 1$ if and only if F^d is well-covered for all $d \geq 1$.*

Proof. Since u and v are true twins, contracting the edge uv to a single vertex w yields a graph F isomorphic to the deletion $G - v$. Consequently, G is isomorphic to the clique blow-up of F obtained by replacing the vertex w with a clique K_2 and replacing every other vertex $x \in V(F) \setminus \{w\}$ with K_1. The result then follows directly from Theorem 1.6. □

Proposition 3.6. *Let G be a connected simple graph with $V(G) = \{v_1, \ldots, v_n\}$, and let $r \geq 1$ be an integer. Let $H = G[rK_1, \ldots, rK_1]$ be the r-fold independence blow-up of G. Then $\operatorname{diam}(H) = \max\{\operatorname{diam}(G), 2\}$ for $r \geq 2$, and H^d is well-covered for all $d \geq 1$ if and only if G^d is well-covered for all $d \geq 1$.*

Proof. Let $Q_1, \ldots, Q_n$ denote the independent sets of size r in H corresponding to the vertices $v_1, \ldots, v_n$ of G, respectively, and let $\pi : V(H) \to V(G)$ be the canonical projection mapping each $x \in Q_i$ to v_i. Two distinct vertices $x, y \in V(H)$ are adjacent in H if and only if $\pi(x) \neq \pi(y)$ and $\{\pi(x), \pi(y)\} \in E(G)$.

First, we analyze the distance function d_H on $V(H)$:

(1) If $x, y \in Q_i$ with $x \neq y$ (which can occur when $r \geq 2$), then x and y are non-adjacent false twins in H. Since G is connected, v_i has at least one neighbor $v_j \in V(G)$. Any $z \in Q_j$ is adjacent to both x and y in H, so (x, z, y) is a path of length 2 in H. Thus $d_H(x, y) = 2$.
(2) If $x \in Q_i$ and $y \in Q_j$ with $i \neq j$, any shortest path between v_i and v_j in G lifts to a path of the same length between x and y in H, and any path between x and y in H projects to a walk of equal length between v_i and v_j in G. Thus $d_H(x, y) = d_G(v_i, v_j)$.

Taking the maximum over all pairs $x, y \in V(H)$, it follows that $\operatorname{diam}(H) = \max\{\operatorname{diam}(G), 2\}$ when $r \geq 2$ (and $\operatorname{diam}(H) = \operatorname{diam}(G)$ when $r = 1$).

Next, fix $d \geq 1$. From the distance formula above, two distinct vertices $x, y \in V(H)$ are adjacent in H^d if and only if either $d \geq 2$ and $\pi(x) = \pi(y)$, or $\{\pi(x), \pi(y)\} \in E(G^d)$.

If $d = 1$, each Q_i is an independent set in H, and every maximal independent set I of H contains at most one vertex from each Q_i. Hence, π maps maximal independent sets of H bijectively to maximal independent sets of G, preserving cardinality. If $d \geq 2$, each Q_i becomes a clique in H^d, so any maximal independent set I of H^d contains exactly one vertex from each Q_i for $v_i \in J$, where $J = \pi(I)$ is a maximal independent set of G^d.

In both cases, all maximal independent sets of H^d have equal cardinality if and only if all maximal independent sets of G^d have equal cardinality. Therefore, H^d is well-covered for all $d \geq 1$ if and only if G^d is well-covered for all $d \geq 1$. □

Example 3.7. Consider the following examples of clique and independence blow-ups of C_5:

(1) Let $H_1 = C_5[K_2, K_1, K_1, K_1, K_1]$ be the clique blow-up of C_5 obtained by replacing a single vertex with a clique K_2.
(2) Let $H_2 = C_5[2K_1, 2K_1, 2K_1, 2K_1, 2K_1]$ be the 2-fold independence blow-up of C_5 obtained by replacing every vertex with an independent set $2K_1$.

By Theorem 1.6 and Proposition 3.6, both H_1^d and H_2^d are well-covered for all $d \geq 1$, as C_5^d is well-covered for all $d \geq 1$.

The following example illustrates that there exist graphs of diameter 3 whose powers are all well-covered.

Example 3.8. Let $G = C_7$, which has diameter 3. Then G^d is well-covered for all $d \geq 1$. Similarly, if $G = C_5 \square K_2$ (the pentagonal prism), then $\operatorname{diam}(G) = 3$ and G^d is well-covered for all $d \geq 1$.



H_1 H_2

FIGURE 1. The clique blow-up H_1 and the 2-fold independence blow-up H_2 of C_5.

By Corollary 3.5, when investigating graphs G such that G^d is well-covered for all $d \geq 1$, we may assume without loss of generality that G has no true twins. We do not know of any examples of such graphs with diameter at least 4. On the other hand, every well-covered graph of diameter 2 automatically satisfies this condition. This leads us to pose the following question.

Question 3.9. Let G be a simple connected graph such that G^d is well-covered for all $d \geq 1$. Is it true that $\mathrm{diam}(G) \leq 3$? Furthermore, if $\mathrm{diam}(G) = 3$, are there only finitely many such graphs without true twins?

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Authors' contributions.

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